A SOURCE-ANCHORED ACKERMANN COMPARISON FOR THE SEIDEL–SHARIR TOP-DOWN PATH-COMPRESSION HIERARCHY

PATH-COMPRESSION PROOF-DIGESTION PACKET

Abstract. This exposition packages the proof that the Seidel–Sharir top-down J hierarchy implies an inverse-Ackermann bound under the packet normalization. The main comparison is

$$R_{z+1}(Q) \geq A(z, 4Q) \qquad (z \geq 1,\ Q \geq 1),$$

with constants $c = 1$, $C = 1$, and $D = 4$ in the comparison template $R_{cz+C}(Q) \geq A(z, DQ)$. The note is a source-fidelity and threshold-inversion exposition. It is not a new union-find bound, and it is not a replacement for the full Seidel–Sharir recurrence proof.

Contents

1. Introduction

Tarjan's Dagstuhl prompt asks for a simple direct proof that the Seidel–Sharir top-down J hierarchy for path compression implies an inverse-Ackermann bound using a classical Ackermann definition; see the problem anchor in the Dagstuhl report[1] and the original Seidel–Sharir paper [2, 1]. The Dagstuhl

Date: May 17, 2026.

[1]Dagstuhl Report 15(5), Seminar 25191, Section 5.3, asks: "Seidel and Sharir gave a proof of the inverse-Ackermann-function upper bound for path compression based on a beautiful top-down recurrence. The analysis does not use Ackermann's function explicitly, but uses a related function they call 'J.' Problem: Give a simple, direct proof of the inverse-Ackermann function bound using their top-down recurrence and the classical definition of Ackermann's function."

§5.3 prompt quotation in the footnote was checked against the public Dagstuhl Report 15(5), §5.3.

This note proves that comparison under the packet normalization. It uses the source J hierarchy and the packet Ackermann function to prove

$$R_{z+1}(Q) \geq A(z, 4Q) \qquad (z \geq 1,\ Q \geq 1).$$

Thus the packet comparison constants are explicit:

$$c = 1, \qquad C = 1, \qquad D = 4.$$

The role of this document is deliberately narrow. It does not claim a new union-find bound, does not reprove the full Seidel–Sharir top-down recurrence, and does not replace that recurrence with a bottom-up Tarjan-potential argument. The only recurrence used here is the source-anchored consequence, where $f(m, n, r)$ denotes the path-compression cost function in the Seidel–Sharir recurrence [1, p. 9],

$$f(m, n, r) \leq km + 2nJ_k(r).$$

2. Source Recurrence and Conventions

All logarithms are base 2. The source definitions and recurrence quoted in this section are from Seidel–Sharir [1, pp. 7–9]. For an integer-valued function g with $g(r) < r$ for positive r, the source defines

$$g^*(r) = \begin{cases} 0, & r \leq 1, \\ 1 + g^*(g(r)), & r > 1, \end{cases}$$

and

$$g^\diamond(r) = \begin{cases} g(r), & g(r) \leq 1, \\ 1 + g^\diamond(\lceil \log_2 g(r) \rceil), & g(r) > 1. \end{cases}$$

The source shifting lemma, with the $2n$ coefficient used below, is Lemma 5 of Seidel–Sharir [1, pp. 7–8].

Lemma 2.1 (Source-correct shifting lemma)**.** *If*

$$f(m, n, r) \leq km + 2ng(r),$$

then

$$f(m, n, r) \leq (k + 1)m + 2ng^\diamond(r).$$

The source hierarchy is Corollary 6 of Seidel–Sharir [1, p. 9]:

$$J_0(r) = \left\lceil \frac{r-1}{2} \right\rceil, \qquad J_{k+1} = J_k^\diamond.$$

Equivalently,

$$J_{k+1}(r) = \begin{cases} J_k(r), & J_k(r) \leq 1, \\ 1 + J_{k+1}(\lceil \log_2 J_k(r) \rceil), & J_k(r) > 1. \end{cases}$$

The same corollary gives the source recurrence consequence used in the cost bound:

$$f(m, n, r) \le km + 2nJ_k(r).$$

3. Threshold Inversion

For $t \ge 0$, define the threshold inverse

$$R_k(t) = \max\{r \ge 0 : J_k(r) \le t\}.$$

The following package supplies the finite-threshold and unboundedness facts needed for this definition.

Lemma 3.1 (Basic J package)**.** *For every $k \ge 0$:*

(1) *J_k is integer-valued, nonnegative, nondecreasing, and unbounded on $\mathbb{N}$.*
(2) *$J_k(0) = 0$ and $J_k(r) < r$ for every $r > 1$.*
(3) *$J_{k+1}(r) \le J_k(r)$ for every $r \ge 0$.*
(4) *$R_k(t)$ is finite for every finite $t \ge 0$.*
(5) *$R_k(t)$ is nondecreasing in t, and $R_{k+1}(t) \ge R_k(t)$.*

Proof. For $k = 0$, the formula $J_0(r) = \lceil (r-1)/2 \rceil$ gives the asserted properties directly. For the induction step, let $h = g^\diamond$ where g is integer-valued, nonnegative, nondecreasing, unbounded, satisfies $g(0) = 0$, and $g(r) < r$ for $r > 1$. Since

$$\lceil \log_2 x \rceil \le x - 1 \qquad (x \ge 2),$$

the recursive branch of h calls h only on smaller arguments. A strong induction on r gives $h(r) \le g(r)$ for all r. Indeed, if $g(r) \le 1$, then $h(r) = g(r)$. If $g(r) = 2$, then $h(r) = 1 + h(1)$. Monotonicity of g and $g(2) < 2$ give $g(1) \le 1$, so $h(1) = g(1) \le 1$ and $h(r) \le 2 = g(r)$. If $g(r) \ge 3$, set $a = \lceil \log_2 g(r) \rceil$. Then $a \le g(r) - 1 \le r - 2$, so the induction hypothesis applies. Hence

$$h(r) = 1 + h(a) \le 1 + g(a) \le 1 + g(g(r) - 1).$$

Since $g(r) - 1 \ge 2$ and $g(s) < s$ for $s > 1$, $g(g(r) - 1) \le g(r) - 2$, and therefore $h(r) \le g(r) - 1 < g(r)$. Thus $h(r) \le g(r)$ for all r, hence $h(r) < r$ for $r > 1$ and $h(0) = 0$.

Note that $h \ge 0$: if $g(r) \le 1$, then $h(r) = g(r) \ge 0$; otherwise $h(r) = 1 + h(\lceil \log_2 g(r) \rceil) \ge 1$.

For monotonicity, we prove by strong induction on s that $r \le s$ implies $h(r) \le h(s)$. Let $r \le s$ and put $x = g(r)$ and $y = g(s)$. Since g is nondecreasing, $x \le y$. If $y \le 1$, then $h(r) = x \le y = h(s)$. If $x \le 1 < y$, then

$$h(r) = x \le 1 \le 1 + h(\lceil \log_2 y \rceil) = h(s),$$

using nonnegativity of h. If $x > 1$, put $u = \lceil \log_2 x \rceil$ and $v = \lceil \log_2 y \rceil$. The ceiling logarithm is nondecreasing, so $u \le v$, and $v \le y - 1 < s$. The strong induction hypothesis gives $h(u) \le h(v)$, whence $h(r) = 1 + h(u) \le 1 + h(v) = h(s)$.

To see unboundedness, we induct on $M \ge 0$. The base case $M = 0$ holds because $h(0) = 0$. For the inductive step, suppose $h(u) \ge M$. Since g is unbounded, choose r with $g(r) > 2^u$; then $\lceil \log_2 g(r) \rceil > u$, and monotonicity gives

$$h(r) = 1 + h(\lceil \log_2 g(r) \rceil) \ge M + 1.$$

Thus $g^\diamond$ preserves the stated properties, and induction gives the claims for every J_k.

Since each J_k is nondecreasing and unbounded, the set $\{r \ge 0 : J_k(r) \le t\}$ is nonempty and finite for finite $t \ge 0$. This proves finiteness of $R_k(t)$. Monotonicity in t is immediate from inclusion of threshold sets, and $J_{k+1} \le J_k$ gives $R_{k+1}(t) \ge R_k(t)$. $\square$

Lemma 3.2 (Exact base inverse)**.** *For every $t \ge 0$,*

$$R_0(t) = 2t + 1.$$

Proof. The inequality $J_0(r) \le t$ is equivalent to $\lceil (r-1)/2 \rceil \le t$, hence to $r \le 2t + 1$. $\square$

Remark 3.3. For every $k \ge 0$, $R_k(0) = 1$. Indeed, $J_k(0) = J_k(1) = 0$ and $J_k(2) = 1$ by induction from J_0 and the definition of $g^\diamond$.

Lemma 3.4 (Threshold step)**.** *Let B, t be nonnegative integers. If*

$$r \le R_k(2^B) \qquad \text{and} \qquad B \le R_{k+1}(t),$$

then

$$J_{k+1}(r) \le t + 1.$$

Consequently,

$$R_{k+1}(t+1) \ge R_k(2^{R_{k+1}(t)}).$$

Proof. The first hypothesis gives $J_k(r) \le 2^B$. If $J_k(r) \le 1$, then $J_{k+1}(r) = J_k(r) \le 1 \le t + 1$. If $J_k(r) > 1$, then

$$J_{k+1}(r) = 1 + J_{k+1}(\lceil \log_2 J_k(r) \rceil),$$

and $\lceil \log_2 J_k(r) \rceil \le B$. Since J_{k+1} is nondecreasing and $B \le R_{k+1}(t)$,

$$J_{k+1}(\lceil \log_2 J_k(r) \rceil) \le J_{k+1}(B) \le t.$$

Thus $J_{k+1}(r) \le t + 1$. Taking $B = R_{k+1}(t)$ gives the displayed threshold recurrence. $\square$

Lemma 3.5 (Threshold jump)**.** *For every $k \ge 0$ and every $Q \ge 2$,*

$$R_{k+1}(Q) \ge R_k(4Q).$$

Proof. Apply Lemma 3.4 with $t = Q - 1$:

$$R_{k+1}(Q) \ge R_k(2^{R_{k+1}(Q-1)}).$$

The basic package and the exact base inverse give $R_{k+1}(Q-1) \ge R_0(Q-1) = 2Q - 1$. For $Q \ge 2$, $2^{2Q-1} \ge 4Q$, and monotonicity of R_k finishes the proof. $\square$

4. Ackermann Comparison

The packet Ackermann function is defined exactly as follows:

$$A(0,x) = 2x,$$

and for $i \geq 1$,

$$A(i,0) = 0, \qquad A(i,1) = 2, \qquad A(i,x) = A(i-1, A(i,x-1)) \quad (x \geq 2).$$

Lemma 4.1 (Ackermann monotonicity and row domination)**.** *For the packet Ackermann function:*

(1) *for every $i \geq 0$, $A(i,x)$ is nondecreasing in x;*
(2) *for every $i \geq 1$ and $x \geq 1$, $A(i,x) \geq 2x$;*
(3) *for every $i \geq 0$ and $x \geq 1$, $A(i+1,x) \geq A(i,x)$.*

Proof. For $i = 0$, monotonicity follows from $A(0,x) = 2x$. For $i \geq 1$, monotonicity follows inductively from $A(i,x+1) = A(i-1, A(i,x))$ and the lower-row monotonicity.

The growth bound is by induction on x. At $x = 1$ it says $2 = 2x$. For $x \geq 2$,

$$A(i,x) = A(i-1, A(i,x-1)).$$

If $i = 1$, this equals $2A(1,x-1)$; if $i \geq 2$, the lower-row growth bound gives at least $2A(i,x-1)$. Hence $A(i,x) \geq 2x$.

For row domination, equality holds at $x = 1$. For $x \geq 2$,

$$A(i+1,x) = A(i, A(i+1,x-1)).$$

The growth bound gives $A(i+1,x-1) \geq 2(x-1) \geq x$, and monotonicity in row i gives $A(i+1,x) \geq A(i,x)$. □

Corollary 4.2. For every $y \geq 1$, $A(1,y) = 2^y$.

Proof. The claim follows by induction on y. The base case is $A(1,1) = 2$. For $y \geq 2$,

$$A(1,y) = A(0, A(1,y-1)) = 2A(1,y-1).$$

□

Lemma 4.3 (Row-domination invariant)**.** *For every $j \geq 1$ and $x \geq 1$,*

$$R_j(x) \geq A(j+1,x).$$

Proof. We prove this by induction on j, with an inner induction on x.

For $j = 1$ and $x = 1$, Lemma 3.4 with $k = 0, t = 0$ gives

$$R_1(1) \geq R_0(2^{R_1(0)}).$$

Since $R_1(0) \geq R_0(0) = 1$, this gives $R_1(1) \geq R_0(2) = 5 \geq 2 = A(2,1)$. If $R_1(x) \geq A(2,x)$, then

$$R_1(x+1) \geq R_0(2^{R_1(x)}) = 2 \cdot 2^{R_1(x)} + 1 \geq 2^{A(2,x)} = A(2,x+1).$$

Here the last equality uses Corollary 4.2.

Now assume $R_j(y) \ge A(j+1, y)$ for all $y \ge 1$. We prove the claim for $j+1$. At $x = 1$, $R_{j+1}(1) \ge R_0(1) = 3 \ge 2 = A(j+2, 1)$. If $R_{j+1}(x) \ge A(j+2, x)$, then Lemma 3.4 gives

$$R_{j+1}(x+1) \ge R_j(2^{R_{j+1}(x)}).$$

By Corollary 4.2 and the inner inductive hypothesis,

$$2^{R_{j+1}(x)} = A(1, R_{j+1}(x)) \ge A(1, A(j+2, x)) \ge A(j+2, x),$$

where the last inequality uses Lemma 4.1(2). The outer induction hypothesis applies to the argument $2^{R_{j+1}(x)}$, and Ackermann monotonicity in the second argument then gives

$$R_j(2^{R_{j+1}(x)}) \ge A(j+1, 2^{R_{j+1}(x)}) \ge A(j+1, A(j+2, x)).$$

The last expression is $A(j+2, x+1)$ by the Ackermann recursion. $\square$

Theorem 4.4 (Main comparison)**.** *For all integers $z \ge 1$ and $Q \ge 1$,*

$$R_{z+1}(Q) \ge A(z, 4Q).$$

Consequently,

$$A(z, 4Q) > r \implies J_{z+1}(r) \le Q.$$

Proof. First suppose $Q \ge 2$. By Lemma 3.5, $R_{z+1}(Q) \ge R_z(4Q)$. Since $z \ge 1$, Lemma 4.3 gives $R_z(4Q) \ge A(z+1, 4Q)$. Row domination from Lemma 4.1 gives $A(z+1, 4Q) \ge A(z, 4Q)$.

For $Q = 1, z = 1$, Lemma 3.4 with $k = 1, t = 0$ gives $R_2(1) \ge R_1(2^{R_2(0)})$. Since $R_2(0) \ge R_0(0) = 1$, monotonicity gives $R_2(1) \ge R_1(2)$. The same lemma with $k = 0, t = 0$ gives $R_1(1) \ge R_0(2^{R_1(0)}) \ge R_0(2) = 5$. Applying it again with $k = 0, t = 1$ gives $R_1(2) \ge R_0(2^{R_1(1)}) \ge R_0(2^5) = 65 \ge 16 = A(1, 4)$.

For $Q = 1, z \ge 2$, Lemma 3.4 with $k = z, t = 0$ gives $R_{z+1}(1) \ge R_z(2^{R_{z+1}(0)})$. Since $R_{z+1}(0) \ge R_0(0) = 1$, monotonicity gives $R_{z+1}(1) \ge R_z(2)$. Applying Lemma 3.5 with level $z-1$ and $Q = 2$ gives $R_z(2) \ge R_{z-1}(8)$. Since $z - 1 \ge 1$, Lemma 4.3 gives $R_{z-1}(8) \ge A(z, 8) \ge A(z, 4)$.

Thus $R_{z+1}(Q) \ge A(z, 4Q)$ in all cases. If $r \ge 0$ is an integer and $A(z, 4Q) > r$, then $r \le A(z, 4Q) - 1 \le R_{z+1}(Q)$, so $J_{z+1}(r) \le Q$ by the definition of R_{z+1}. $\square$

5. Path-Compression Consequence

Define

$$L(n) = \lceil \log_2 \max(n, 2) \rceil, \qquad Q(m, n) = \left\lceil 1 + \frac{m}{n} \right\rceil.$$

Define

$$\alpha_Q(m, n) = \min\{z \ge 1 : A(z, 4Q(m, n)) > L(n)\},$$
$$\alpha_J^Q(m, n) = \min\{k \ge 0 : J_k(L(n)) \le Q(m, n)\},$$

and

$$\alpha_J^S(m, n) = \min\left\{k \ge 0 : J_k(L(n)) \le 1 + \frac{m}{n}\right\}.$$

Lemma 5.1 (Existence of α_Q)**.** *For positive integers m, n, the minimum defining $\alpha_Q(m,n)$ exists.*

Proof. Since $Q(m,n) \geq 1$, it suffices to prove that $A(z,4)$ is unbounded in z. Let $a_z = A(z,4)$. Then $a_1 = 16$, and for $z \geq 1$,

$$A(z+1,2) = 4, \qquad A(z+1,3) = A(z,4) = a_z,$$

so $a_{z+1} = A(z+1,4) = A(z,a_z)$. By Lemma 4.1, $A(z,a_z) \geq 2a_z$. Hence a_z is unbounded, and monotonicity in the second argument gives $A(z,4Q(m,n)) \geq A(z,4)$. $\square$

Theorem 5.2 (Packet threshold comparison)**.** *For all positive integers m, n,*

$$\alpha_J^Q(m,n) \leq \alpha_Q(m,n) + 1.$$

Proof. Let $z = \alpha_Q(m,n)$, $Q = Q(m,n)$, and $L = L(n)$. By definition, $A(z,4Q) > L$. Theorem 4.4 gives $R_{z+1}(Q) \geq A(z,4Q) > L$. Since L is an integer, $L \leq R_{z+1}(Q)$, so $J_{z+1}(L) \leq Q$. Therefore $\alpha_J^Q(m,n) \leq z+1 = \alpha_Q(m,n)+1$. $\square$

Lemma 5.3 (Ackermann buffer)**.** *For $z \geq 1$ and integer $p \geq 1$,*

$$A(z+1,4p) \geq A(z,4p+4).$$

Proof. Since $4p \geq 2$,

$$A(z+1,4p) = A(z, A(z+1,4p-1)).$$

If $p = 1$, then $A(z+1,3) = A(z,4) \geq A(1,4) = 16 \geq 8 = 4p+4$, using row domination for $z \geq 1$. If $p \geq 2$, then Lemma 4.1 gives $A(z+1,4p-1) \geq 2(4p-1) = 8p-2 \geq 4p+4$. Monotonicity of $A(z,-)$ finishes the proof. $\square$

Theorem 5.4 (Source-faithful real threshold)**.** *For all positive integers m, n,*

$$\alpha_J^S(m,n) \leq \alpha_Q(m,n) + 2.$$

If $1 + m/n$ is integral, then

$$\alpha_J^S(m,n) \leq \alpha_Q(m,n) + 1.$$

Proof. Let $s = 1 + m/n$, $Q = \lceil s \rceil$, $L = L(n)$, and $z = \alpha_Q(m,n)$. Then $A(z,4Q) > L$.

If s is integral, then $Q = s$. Theorem 4.4 gives $R_{z+1}(s) \geq A(z,4Q) > L$, hence $J_{z+1}(L) \leq s$. Thus $\alpha_J^S(m,n) \leq z+1$.

If s is not integral, put $p = \lfloor s \rfloor$. Since $m, n \geq 1$, $p \geq 1$, and $Q = p+1$. Thus $A(z,4p+4) > L$. By Lemma 5.3, $A(z+1,4p) > L$. Theorem 4.4 gives $R_{z+2}(p) \geq A(z+1,4p) > L$, so $J_{z+2}(L) \leq p < s$. Therefore $\alpha_J^S(m,n) \leq z+2$. $\square$

Theorem 5.5 (Cost consequence)**.** *Using the source recurrence*

$$f(m,n,r) \leq km + 2nJ_k(r),$$

one obtains

$$f(m,n,L(n)) \leq (\alpha_Q(m,n)+3)m + 4n.$$

Consequently,

$$f(m,n,L(n)) = O(m\alpha_Q(m,n)+n).$$

Proof. Let $z = \alpha_Q(m,n)$, $Q = Q(m,n)$, and $L = L(n)$. By Theorem 5.2, $J_{z+1}(L) \le Q$. Apply the source recurrence with $k = z+1$:

$$f(m,n,L) \le (z+1)m + 2nJ_{z+1}(L) \le (z+1)m + 2nQ.$$

Since

$$Q = \left\lceil 1 + \frac{m}{n} \right\rceil \le 2 + \frac{m}{n},$$

we have $2nQ \le 4n + 2m$. Therefore

$$f(m,n,L(n)) \le (\alpha_Q(m,n)+3)m + 4n.$$

The big-O statement follows immediately. □

6. Normalization and Source-Fidelity Comments

The packet integer threshold

$$Q(m,n) = \left\lceil 1 + \frac{m}{n} \right\rceil$$

is not an original source threshold. It is a normalization that lets the integer-valued inverse $R_k(t)$ compare directly with the source real threshold $1+m/n$ [1, p. 9]. The exact real threshold is handled by Theorem 5.4; the nonintegral case costs one extra level.

The source page 10 comparison with Tarjan's α_T [1, p. 10] is context, not a proof dependency here. The proof above uses only the source J hierarchy, the source shifting lemma and recurrence, the packet Ackermann normalization, and the packet thresholds $L(n)$ and $Q(m,n)$.

The main hazards avoided are:

- the wrong shifting-lemma factor: the source-correct premise is $f(m,n,r) \le km + 2ng(r)$;
- a silent replacement of the source threshold $1 + m/n$ by $2m/n$;
- drift into a bottom-up Tarjan-potential proof;
- fake equivalence wording: Theorem 4.4 gives the displayed implication, not an unwarranted equivalence.

Appendix A. Dependency Table and DAG

Stage	Input	Output
Source definitions	g^*, $g^\diamond$, J_0, $J_{k+1} = J_k^\diamond$	The J hierarchy
Source recurrence	Shifting lemma with $2ng(r)$ premise	$f(m,n,r) \le km + 2nJ_k(r)$
Basic J lemmas	Diamond preservation	J_k monotone, unbounded, $J_k(r) < r$, finite R_k
Threshold recurrence	$J_{k+1} = J_k^\diamond$	$R_{k+1}(t+1) \ge R_k(2^{R_{k+1}(t)})$
Ackermann domination	Threshold recurrence plus $R_0(t) = 2t+1$	$R_j(x) \ge A(j+1,x)$
Main comparison	Threshold jump and row domination	$R_{z+1}(Q) \ge A(z,4Q)$
Alpha translation	$A(z,4Q) > L(n)$	$\alpha_J^Q \le \alpha_Q + 1$, $\alpha_J^S \le \alpha_Q + 2$
Cost bound	$f \le km + 2nJ_k$	$f(m,n,L(n)) \le (\alpha_Q+3)m + 4n$

A compact text form of the dependency DAG is:

```
source definitions
  -> J hierarchy
  -> basic J lemmas
  -> threshold recurrence
  -> Ackermann domination
  -> R/Ackermann comparison
  -> alpha consequences
  -> cost bound

source shifting lemma
  -> recurrence f(m,n,r) <= k m + 2n J_k(r)
  -> cost bound

packet normalizations L(n), Q(m,n)
  -> alpha consequences
  -> cost bound
```

Appendix B. Finite Sanity Check

The script `scripts/sanity_check_j_thresholds.py` is a finite off-by-one sanity check. It computes small values of the packet J, R, and capped A definitions, checks monotonicity in small ranges, checks small instances of the threshold step, and checks the smallest published comparison instance $R_2(1) \ge A(1,4)$. It is not a proof and is not used as a dependency in the argument above.

References

[1] Raimund Seidel and Micha Sharir. Top-down analysis of path compression. *SIAM Journal on Computing*, 34(3):515–525, 2005. DOI: https://doi.org/10.1137/S0097539703439088; author PDF: https://www.math.tau.ac.il/~michas/ufind.pdf.

[2] Robert Endre Tarjan. Top-down analysis of path compression. Open problem in *Adaptive and Scalable Data Structures*, Dagstuhl Reports 15(5), Report from Dagstuhl Seminar 25191, 2025. Report editors: Michael A. Bender, John Iacono, László Kozma, and Eva Rotenberg. Seminar held May 4–9, 2025; Section 5.3. DOI: https://doi.org/10.4230/DagRep.15.5.1; URL: https://drops.dagstuhl.de/storage/04dagstuhl-reports/volume15/issue05/25191/html/DagRep.15.5.1/DagRep.15.5.1.html.